

Ketan Suresh Shelke

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Education

Bachelor of Engineering (B.E.) in Computer Engineering

Don Bosco Institute of Technology, Mumbai

Mumbai University

Expected Year of Graduation: 2028

Aggregate CGPA: 9.24 / 10 (Upto Semester 3 and rounded to 2 decimals)

- **Current Semester:** 4
- **Major Coursework Undertaken:** Data Structures, Analysis of Algorithms, Database Management Systems, Object Oriented Programming (C++), Computer Architecture, Discrete Mathematics, Python Programming, Applied Mathematics, C Programming.

Projects

Sole Purpose: Footwear E-commerce Platform (Award Winning Project)

- **Full-Stack Development:** Engineered an award-winning, responsive e-commerce application using **PHP** and **MySQL** for robust backend architecture, integrated with **HTML5**, **CSS3**, and **JavaScript** for a dynamic frontend.
- **Custom Algorithms:** Designed and implemented a proprietary "Foot Quiz" recommendation engine and a "Smart Size Converter" algorithm, delivering highly personalized product matching based on foot health data.
- **API & System Performance:** Developed asynchronous **AJAX** endpoints to enable real-time Cart and Wishlist updates without page reloads, significantly improving UI/UX and user engagement.
- **Database & Security:** Architected scalable relational database schemas for efficient inventory management; implemented secure user authentication, session handling, and full **CRUD** operations to ensure data privacy.

ReverseOTP: Power Shift in Welfare Distribution System | HackSpark 2.0 Hackathon Feb 20th -21st, 2026

- **Objective:** Developed an **offline-first, decentralized verification system** for the Public Distribution System (PDS) to eliminate middleman corruption and black-market diversion.
- **Technical Implementation (System Logic Lead):** Engineered the core verification logic using **Node.js** and **Express.js**, implementing **SHA-256 hash-chaining** to create tamper-evident local transaction records.
- **Key Features:**
 - Designed a **User-initiated verification flow** where beneficiaries receive a **Unique Transaction ID (UTID)** via **USSD/SMS**, shifting power from the shopkeeper to the citizen.
 - Implemented a **dual-database architecture** using **SQLite** for offline-first local ledgers and **MongoDB** for centralized cloud synchronization and auditing.
 - Built an **auto-flagging mechanism** that triggers government inspection alerts based on beneficiary-reported mismatch complaints.

Technology Stack

Languages: C++, C, Java, Python, PHP, JavaScript, SQL, Assembly (8086)

Web Technologies: HTML, CSS

Databases: MySQL

Coding and Project Management Tools: Microsoft VSCode, GitHub, Git

Coding Profiles:  LeetCode

Honors & Achievements

- **Runner-up, 'InnoQuest' Project Competition (2025):** Secured 2nd Rank among Second Year Engineering teams for the project "Sole Purpose," awarded for excellence in Full Stack implementation. 
- **Top 40 Finalist, HackNiche 4.0 (2026):** Selected from 892 competing teams to build "AaharIQ," an AI-powered hospitality strategy tool in 24 hours. Secured an exclusive live demo request from the event sponsor for excellence in data-driven market analysis using React 19 and Django REST Framework.

Positions of Responsibility & Volunteering

- 1. Puzzle Team Member** - Domain: The Mathematics Club of D.B.I.T. Aug 2025 – Present
 - Collaborated with the technical team to curate and design logic-based puzzles for club events and student competitions.
- 2. Peer Mentor** – Computer Engineering Department, D.B.I.T. Nov 2025 - Present
 - **Student Mentorship:** Selected to guide and mentor First Year Engineering (FE) students, assisting them with academic planning and facilitating their transition into college life.
- 3. Student Volunteer** – National Service Scheme (NSS), D.B.I.T.
 - **Community Service:** Actively participated in social service campaigns and community development initiatives organized by the college unit.